WORKED SOLUTION

STEP II 2013, Question 4

Pure • Coordinate geometry

STEP II 2013, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s2-q4

The key idea

The midpoint of a chord can be found without ever finding the chord's ends: the x-coordinates of P and Q are the roots of a quadratic, and their average is minus half the ratio of the first two coefficients. After that, each locus is a matter of eliminating a parameter. Geometry explains every answer: the line from the centre of a circle to the midpoint of a chord is perpendicular to the chord.

Solution

The coordinates of M

The line is $y = b(x - a)$. Substituting into $x^2 + y^2 = 1$ gives

$$(1 + b^2)x^2 - 2ab^2x + a^2b^2 - 1 = 0.$$

Its roots are the x-coordinates x_P, x_Q of P and Q, so by the sum of the roots

$$x_M = \tfrac{1}{2}(x_P + x_Q) = \frac{ab^2}{1 + b^2}.$$

Since M lies on the line, $y_M = b(x_M - a) = b \cdot \dfrac{ab^2 - a - ab^2}{1 + b^2}$, so

$$M = \left(\frac{ab^2}{1 + b^2},\ -\frac{ab}{1 + b^2} \right).$$

For P and Q to exist and be distinct the discriminant must be positive:

$$4a^2b^4 - 4(1 + b^2)(a^2b^2 - 1) = 4(1 + b^2 - a^2b^2) > 0 \iff a^2b^2 < 1 + b^2.$$

(This says the distance $|ab|/\sqrt{1 + b^2}$ from O to the line is less than 1.) We need this condition in each part.

Part (i)

With $b > 0$ fixed,

$$-by_M = -b \cdot \left(-\frac{ab}{1+b^2}\right) = \frac{ab^2}{1+b^2} = x_M,$$

so $x = -by$ at every point of the locus, as required. This is the line through O with gradient $-1/b$, perpendicular to the chord, just as the key idea predicts.

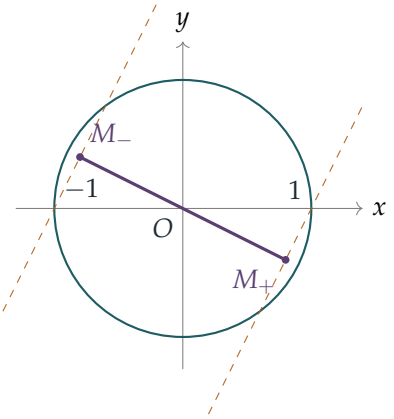
If $-1 \le a \le 1$ then $a^2b^2 \le b^2 < 1 + b^2$, so the line always meets the circle in two points and M is defined for every such a. Both coordinates of M are a times a constant, so as a runs over $[-1, 1]$ the point M moves steadily along the line $x = -by$ from the point given by $a = -1$ to the point given by $a = 1$. The locus is therefore the line segment joining

$$M_\pm = \pm\left(\frac{b^2}{1+b^2},\ -\frac{b}{1+b^2}\right),$$

whose length is

$$2\,\frac{\sqrt{b^4+b^2}}{1+b^2} = \frac{2b\sqrt{1+b^2}}{1+b^2} = \frac{2b}{(1+b^2)^{1/2}},$$

using $b > 0$ to take the square root of b^2. This is the required length. The segment is centred at O; its end M_+ is the midpoint of the chord through $(1, 0)$, and M_- that of the chord through $(-1, 0)$. The sketch shows the case $b = 2$, where the ends are $\pm(\frac{4}{5}, -\frac{2}{5})$.



The dashed lines are the extreme chords, through $(\pm 1, 0)$ with gradient 2.

Part (ii)(a)

Now a is fixed with $0 < a < 1$. Then $a^2b^2 < b^2 < 1 + b^2$, so every line through $A = (a, 0)$ cuts the circle twice. From the formula for M,

$$x^2 + y^2 = \frac{a^2b^4 + a^2b^2}{(1+b^2)^2} = \frac{a^2b^2}{1+b^2} = a \cdot \frac{ab^2}{1+b^2} = ax.$$

So M lies on

$$x^2 + y^2 - ax = 0, \quad \text{that is} \quad (x - \tfrac{a}{2})^2 + y^2 = (\tfrac{a}{2})^2,$$

the circle with centre $(\frac{a}{2}, 0)$ and radius $\frac{a}{2}$, which has OA as a diameter.

Ask yourself Is the whole of this circle traced out?

Not quite. Since $x = ab^2/(1+b^2) = a - a/(1+b^2)$, we always have $0 \leq x < a$, so the point $A = (a, 0)$ itself is never reached; it would need the vertical line through A, which has no gradient. Every other point is reached: putting $b = \tan\theta$ with $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$,

$$M = (a\sin^2\theta,\ -a\sin\theta\cos\theta) = (\tfrac{a}{2} - \tfrac{a}{2}\cos 2\theta,\ -\tfrac{a}{2}\sin 2\theta),$$

and as 2θ runs over $(-\pi, \pi)$ this covers the circle once, except for the point with $2\theta = \pm\pi$, which is A. So the locus is the circle on OA as diameter with the single point A removed. Geometrically: OM is perpendicular to the chord AM, so $\angle OMA$ is a right angle and M lies on the circle with diameter OA (with $M = O$ when $b = 0$). This circle lies inside the unit circle.

Part (ii)(b)

Now $a = 1/b$ with $0 < b \leq 1$. The condition $a^2b^2 < 1 + b^2$ becomes $1 < 1 + b^2$, which holds. The line is $y = b(x - 1/b) = bx - 1$, so it always passes through $B = (0, -1)$, which is on the unit circle; thus B is always one of P, Q. The formula gives

$$M = \left(\frac{b}{1+b^2},\ -\frac{1}{1+b^2}\right), \qquad x^2 + y^2 = \frac{b^2+1}{(1+b^2)^2} = \frac{1}{1+b^2} = -y.$$

So M lies on

$$x^2 + y^2 + y = 0, \quad \text{that is} \quad x^2 + (y + \tfrac{1}{2})^2 = (\tfrac{1}{2})^2,$$

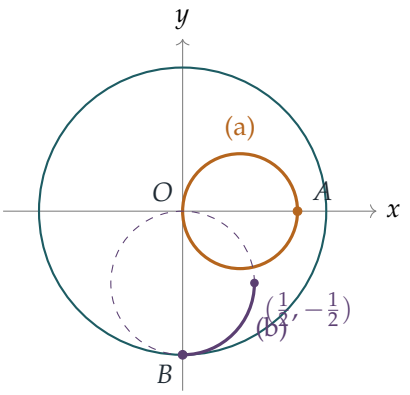
the circle with centre $(0, -\frac{1}{2})$ and radius $\frac{1}{2}$, which has OB as diameter (for the same right-angle reason as in (a)).

This time only part of the circle is traced. With $b = \tan\theta$, $0 < \theta \leq \frac{\pi}{4}$,

$$M = (\tfrac{1}{2}\sin 2\theta,\ -\tfrac{1}{2} - \tfrac{1}{2}\cos 2\theta),$$

and 2θ runs over $(0, \frac{\pi}{2}]$. So the locus is a quarter of the circle: the arc from $B = (0, -1)$, which is not included (it would need $b = 0$), anticlockwise to $(\frac{1}{2}, -\frac{1}{2})$, which is included (from $b = 1$, $a = 1$). On this arc $0 < x \leq \frac{1}{2}$ and $-1 < y \leq -\frac{1}{2}$.

The sketch shows both loci with the unit circle, taking $a = 0.8$ in (a). The open circle marks the missing point A of (a) and the missing end B of (b).



In (b) the solid arc is the locus; the dashed remainder of the small circle is not reached.

Checks

- With $a = 0$ the line passes through O, so the chord is a diameter and M should be O: the formula gives $(0, 0)$.
- In (i) with $b = 1$: the chord through $(1, 0)$ with gradient 1 joins $(1, 0)$ and $(0, -1)$, with midpoint $(\frac{1}{2}, -\frac{1}{2})$, matching M_+. The segment then has length $\sqrt{2} = 2/\sqrt{2}$, as the formula says.
- The same chord is the $b = 1$ end of the arc in (ii)(b), and $(\frac{1}{2}, -\frac{1}{2})$ satisfies $x^2 + y^2 + y = \frac{1}{4} + \frac{1}{4} - \frac{1}{2} = 0$.
- In (ii)(a), $b = 1$ gives $M = (\frac{a}{2}, -\frac{a}{2})$, the lowest point of the circle on OA, as it should be for a chord at 45°.

Common slips

- Finding P and Q explicitly with the quadratic formula. It works, but the sum of the roots gives M in one line.
- Claiming the whole circle in (ii)(a) or (ii)(b). Always check which values the parameter actually takes and which points they give.
- Forgetting to check that the line actually cuts the circle; for (i) and both cases of (ii) it always does, but this needs saying.

Beyond STEP

The fact that midpoints of parallel chords lie on a line through the centre holds for any conic: for an ellipse that line is called a *conjugate diameter*, and it is a first glimpse of the projective idea of poles and polars. The circles in part (ii) are the classical theorem that an angle in a semicircle is a right angle, run backwards. For more on loci and

parametrisation as a problem-solving tool, Siklos, *Advanced Problems in Mathematics*, is a good next step.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.